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Potassium-Argon Age and Paleomagnetism of Diabase Dikes in Liberia: Initiation of Central Atlantic Rifting

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ABSTRACT

Tholeiitic diabase dikes that trend northwest-southeast, parallel to the coastline, are common in northwestern Liberia. K-Ar whole-rock and mineral ages determined from dikes that intrude Precambrian crystalline rocks are discordant and range from 186 to 1,213 m.y. Incremental heating experiments on three neutron-irradiated samples of these rocks give "saddle-shaped" $^{40}Ar/^{39}Ar$ release diagrams that reach minima of less than 300 m.y. at intermediate temperatures and that do not fit a $^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$ isochron. K-Ar ages determined from diabase dikes and sills that intrude Paleozoic sedimentary rocks near the coast are all within the range 173 to 192 m.y. $^{40}Ar/^{39}Ar$ incremental heating data for one of these samples gives a plateau age and a $^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$ isochron age that are concordant with the conventional K-Ar age. The conventional and $^{40}Ar/^{39}Ar$ K-Ar data show that the dikes intruding Precambrian basement rocks contain large and variable amounts of excess ^{40}Ar, whereas the diabase intruding Paleozoic sandstone does not. All of the intrusions are probably earliest Jurassic in age.

Mean paleomagnetic directions in six dikes and sills that intrude sedimentary rocks are nearly parallel to mean paleomagnetic directions in 19 dikes that intrude Precambrian rock, further evidence for contemporaneity. The paleomagnetic pole derived from all 25 diabase units is at lat 68° N., long 242° E., with $\alpha_{95} = 5°$, in close agreement with other Mesozoic paleomagnetic poles from the African continent. A mean paleomagnetic pole for northwest Africa has been calculated using these data and published paleomagnetic directions from 19 other intrusive rock units that have similar radiometric ages in Morocco and Sierra Leone. This pole is compared with another paleomagnetic pole calculated from published data from 16 localities in igneous rocks of latest Triassic to earliest Jurassic age distributed from Nova Scotia to Pennsylvania. The comparison shows that, with the African and North American continents in their present positions, the two poles differ by 44° of arc, but when the continents are restored to the predrift configuration proposed by Bullard and others (1965), the angular difference diminishes to 3°. This coincidence of paleomagnetic poles provides an earliest limit of 180 ± 10 m.y. for the separation of Africa from North America. *Key words: continental drift, sea-floor spreading, opening of Atlantic Ocean, paleomagnetism, Africa, African pole position, Liberia, diabase dikes, potassium-argon, K-Ar, $^{40}Ar/^{39}Ar$, excess argon.*

INTRODUCTION

Parallel dikes of tholeiitic diabase are common in Liberia where they intrude Precambrian basement rocks and, locally near the coast, Paleozoic sedimentary rocks. Their strike is northwest-southeast parallel to the African coast, and geophysical evidence (Behrendt and Wotorson, 1970) indicates that they also occur offshore to the edge of the continental shelf. Behrendt and Wotorson (1970) associated these dikes with the separation of Africa from North and South America.

In December 1968, samples were collected for K-Ar dating and paleomagnetic study. The objectives of the investigation were to provide geochronologic and paleomagnetic data for geologic correlation, to test the hypothesis that the dikes are related to the separation of Africa from North and South America during Mesozoic time, and to obtain an early Mesozoic pole position for West Africa.

GEOLOGIC SETTING

The Precambrian crystalline rocks of Liberia have been subdivided into three provinces of contrasting age, lithology, and structure (Hurley and others, 1971; White and Leo, 1970). The Liberian age province (about 2,700 m.y.) occupies much of the interior of western Liberia (Fig. 1) and consists mostly of granitic gneisses ranging in composition from granite to quartz diorite. Northeast-trending belts of sedimentary rocks, including iron-formation, have been folded into the granitic gneisses and metamorphosed to amphibolite facies. The Pan-African age province (about 550 m.y.) comprises a northwest-trending coastal belt of metasedimentary and mafic rocks that have been refoliated and remetamorphosed to granulite and amphibolite facies and have been intruded by granitic rocks. The Eburnean age province (about 2,000 m.y.) is separated from the Liberian province in eastern Liberia by a belt of refoliated Liberian-type granitic gneisses (not shown in Fig. 1). The southeastern part of the province consists of tightly folded paragneisses of amphibolite facies, amphibolites, migmatites, and granitic intrusive rocks.

Unmetamorphosed sedimentary rocks crop out along the coast of Liberia between Buchanan and the capital city of Monrovia. They include from older to younger, the Paynesville Sandstone, the Farmington River Formation, and the Edina Sandstone (White, 1972). The Paynesville Sandstone, an arenite with a thickness of about 1,000 m on the coast east of Monrovia, lies unconformably on the crystalline basement. Its age is uncertain because fossils are absent, but it is thought to be early or middle Paleozoic because of its stratigraphic position and from comparison with other sedimentary sequences in West Africa. The Paynesville Sandstone is overlain unconformably by wackes and conglomerates of the Farmington River Formation, whose thickness may exceed 1 km (Behrendt and Wotorson, 1970). Pollen and spores indicate that the Farmington River Formation is Cretaceous, probably Albian. The Edina Sandstone (not shown in Fig. 1), of Tertiary(?) age, disconformably overlies the Farmington River Formaton and is only a few meters thick. Unconsolidated Quaternary sediments form a thin veneer along the entire coast of Liberia.

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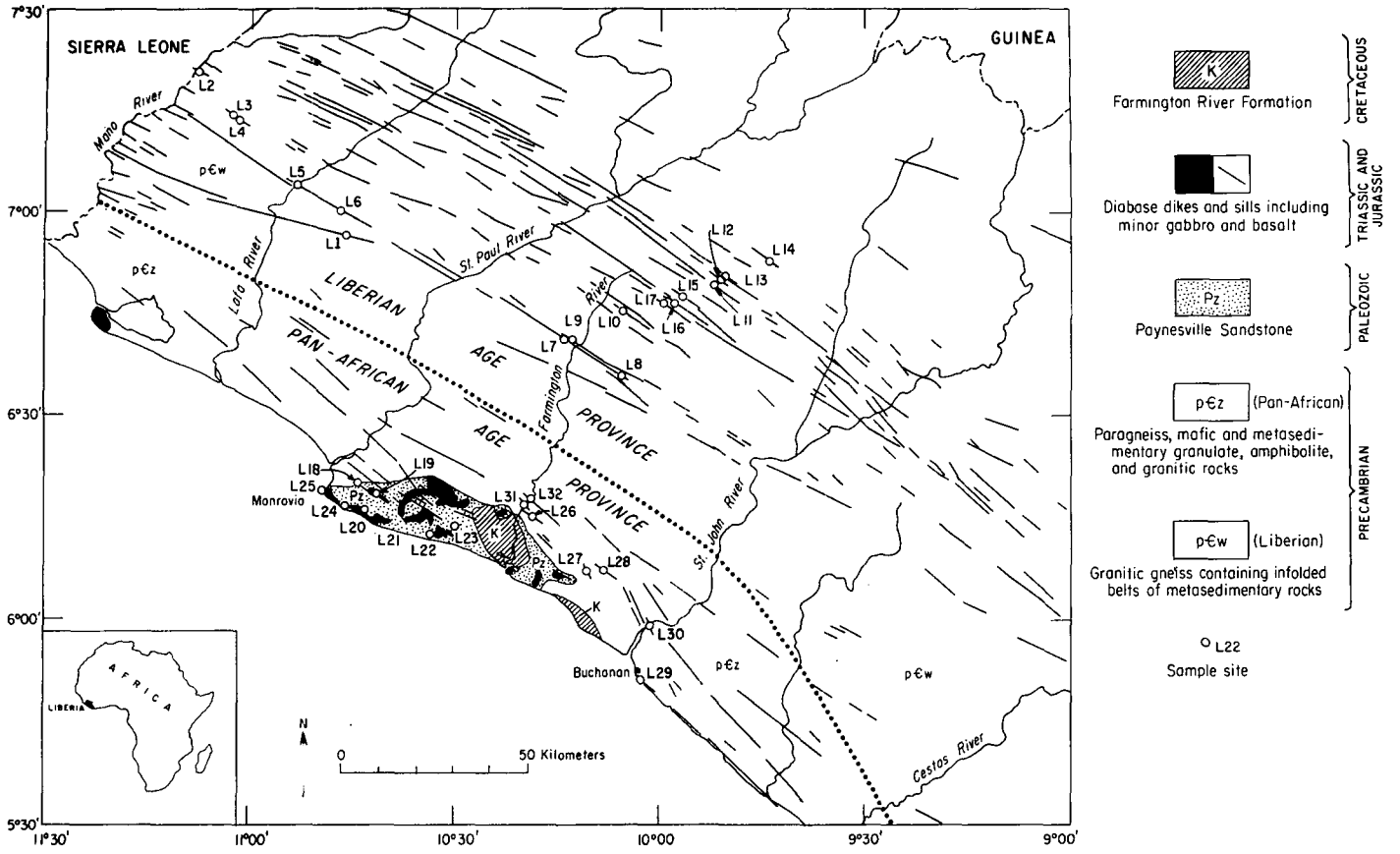


Figure 1. Geologic map of part of western Liberia showing sample locations. Geology after White and Leo (1969) and White (1972), modified from interpretation of aeromagnetic data. Dotted line is approximate boundary between Pan-African age province (ca. 550 m.y.) and Liberian age province (ca. 2,700 m.y.) of Hurley and others (1971).

Dikes of diabase intrude the Precambrian crystalline rocks throughout western Liberia. Three broad geographic zones have been recognized: coastal, central, and northern (White and Leo, 1969; White, 1970; Behrendt and Wotorson, 1972). The coastal and central zones trend northwest-southeast parallel to the coast. Some dikes and sills in the coastal zone intrude the Paynesville Sandstone and are unconformably overlain by the Farmington River Formation; other dikes intrude Precambrian rocks of the Pan-African age province (Fig. 1). An aeromagnetic survey (Behrendt and Wotorson, 1970, 1972) indicates that the dikes also occur beneath sediments on the continental shelf. The central zone, about 90 km inland, consists of abundant dikes that intrude Precambrian rocks of the Liberian age province; it continues northwest into Sierra Leone (Wilson, 1965) but apparently dies out toward the southeast near the border of Liberia and Ivory Coast (Tagini, 1965). The northern zone, which occurs far inland (north of the area of Fig. 1), consists of widely spaced dikes that trend east-west. Dikes of the northern zone not only differ in structural trend from dikes of the coastal and central zones but are mildly metamorphosed and unsuitable for these studies. These dikes probably differ in age and origin from those of the coastal and central zones and will not be discussed further. Except for rough geographic distribution, there is no structural or petrographic evidence that the dikes of the central and coastal zones are of different age or origin; they are considered as a single group throughout most of the following discussion.

Individual dikes vary in length; some are as much as 50 km long but most are less than 10 km long. Thickness ranges from a few centimeters to as much as 100 m but commonly is 15 to 45 m. Most of the dikes are essentially vertical, regardless of the dip of primary structures in the country rock. Even dikes that cut the gently deformed Paynesville Sandstone are vertical, indicating that tectonic rotation since emplacement has been minimal or absent. Several sills are found within the Paynesville Sandstone, and two small flows of tholeiitic basalt overlying the sandstone are thought to be the extrusive equivalent of the diabase (White, 1972). Except for the sills, the trend of the dikes is unrelated to any primary structures in either the sediments or the crystalline basement rocks, and the parallelism of the dikes (Fig. 1) suggests that they are controlled by a through-going fracture system.

The diabase is composed primarily of labradorite and titaniferous augite with minor magnetite, ilmenite, and apatite and, in some dikes, olivine, pigeonite, sulfides, and traces of native copper. Primary graphic intergrowths of accessory quartz and alkali feldspar are common. The diabase is generally fine to medium grained with uniform ophitic to subophitic texture but contains occasional phenocrysts of plagioclase or pyroxene. The thicker dikes are coarse grained. The diabase is generally unaltered where it is unweathered, except for some rocks that have incipient development of sericite in the plagioclase. There is no petrologic or field evidence to suggest that the diabase dikes have been reheated since emplacement. Dikes and sills are petrographically indistinguishable. Chemical analyses of samples from sites L24 (White, 1972), and L29, and two other localities (unpublished) indicate that the diabase is tholeiitic.

SAMPLING AND LABORATORY TECHNIQUES

A total of 316 oriented cores 2.50 cm in diameter was collected using portable equipment (Doell and Cox, 1967a) from 31 sites (shown in Fig. 1). Because of weather and lack of topographic maps, it was necessary to use magnetic orientation exclusively. In all cases the orientations were checked and corrected using a second compass 10 m or more from the outcrop, thus reducing errors

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TABLE 1. ANALYTICAL DATA AND POTASSIUM-ARGON AGES FOR DIABASE DIKES AND SILLS FROM LIBERIA

Site*	Material	k_2O (wt. percent)	$^{40}Ar_{rad}$ (mol/gm)	$\frac{100 \times {}^{40}Ar_{rad}}{{}^{40}Ar_{total}}$	Calculated age† (10^6 years)
		Intrude Precambrian rocks of Liberian age province			
L1	Plagioclase	0.347, 0.354	2.305×10^{-10}	77.5	400 ± 12
L2	Plagioclase	0.337, 0.337	8.47	95.3	1,213 ± 36
L3	Whole rock	0.372, 0.373	5.94	92.0	853 ± 26
	Plagioclase	0.364, 0.365	4.383 4.301	91.0 89.2	671 ± 14
	Plagioclase-L§	0.347, 0.353	3.404 3.454	81.0 86.1	568 ± 12
	Pyroxene	0.091, 0.092	0.763	57.4	493 ± 11
L4	Whole rock	0.401, 0.402	4.427 4.682	94.2 92.1	644 ± 14
L6	Whole rock	0.258, 0.264	5.91	90.7	1,121 ± 34
	Plagioclase	0.280, 0.281	1.984 2.005	80.7 85.3	429 ± 9
	Pyroxene	0.065, 0.066	0.677	55.9	595 ± 23
L7	Whole rock	0.271, 0.276	1.970	79.1	433 ± 13
L8	Whole rock	0.299, 0.300	1.948	77.8	395 ± 12
L9	Whole rock	0.358, 0.360	2.433 2.311	91.8 92.2	401 ± 9
L10	Whole rock	0.795, 0.806	6.95 6.88	91.2 92.5	509 ± 11
L14	Plagioclase	0.308, 0.322	1.103 1.097	80.4 76.5	223 ± 9
		Intrude Paynesville Sandstone			
L19	Plagioclase	0.622, 0.622	1.676 1.652	77.5 60.6	173 ± 4
L20	Plagioclase	0.355, 0.356	1.034 1.078	50.2 55.4	191 ± 4
L21	Plagioclase	0.930, 0.945	2.580 2.578	89.6 87.6	178 ± 4
L22	Plagioclase	0.667, 0.696	1.986	73.2	188 ± 6
L24	Plagioclase	0.305, 0.306	0.832	76.2	176 ± 5
L25	Plagioclase	0.287, 0.288	0.860	76.8	192 ± 6
		Intrude Precambrian rocks of Pan-African age province			
L26	Plagioclase	1.050, 1.067	11.47	86.3	619 ± 19
L27a	Plagioclase	0.657, 0.663	3.138 3.145	82.5 77.3	297 ± 6
b	Whole rock	0.970, 0.970	2.793	89.3	186 ± 6
c	Plagioclase	0.823, 0.823	5.24	13.6	388 ± 25
d	Plagioclase	0.438, 0.449	2.517	85.8	349 ± 10
L28	Plagioclase	0.737, 0.753	4.696 4.807	84.7 66.8	389 ± 8
L29	Plagioclase	0.957, 0.970	2.919 2.869	69.9 73.8	193 ± 4
L30	Plagioclase	1.123, 1.127	7.13	61.2	386 ± 12
L32	Plagioclase	0.628, 0.629	1.266 1.205	91.6 92.8	1,006 ± 21

* The letters a-d indicate different hand samples from the same site.

† Treated with HF (see text).

§ $\lambda_\varepsilon = 0.585 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}, $K^{40}/K_{total} = 1.19 \times 10^{-4}$ mol/mol. The ± figures are estimates of the precision at the 68 percent confidence level (Cox and Dalrymple, 1967).

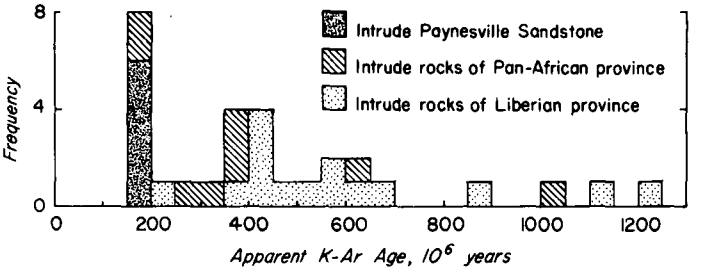


Figure 2. Histogram of K-Ar ages on Liberian diabase dikes and sills. Class interval is 50 m.y.

due to local remagnetization of outcrops by lightning to less than ±1°. One or more hand samples for K-Ar dating were collected from each of the 31 sites and from one additional site (L8). Most sites are in separate dikes or sills and represent separate times of cooling. Possible exceptions are the following three groups of sites: L3 and L4; L5, L6, L7, and L8; and L20 and L24 (see Fig. 1).

Plagioclase was separated for dating where grain size was sufficiently coarse and where plagioclase was fresh. Whole-rock samples were evaluated using the criteria described by Dalrymple and Lanphere (1969) and Mankinen and Dalrymple (1972) to assure that the samples contained no glass, alteration products, secondary mineralization, or any phase that might not have remained a closed system since the rock crystallized.

Conventional argon analyses were done by isotope dilution using a bulb-type ^{38}Ar tracer, a Reynolds-type 4½-in. radius rare-gas mass spectrometer, and techniques and data-reduction procedures described in Dalrymple and Lanphere (1969). Potassium was measured by flame photometry using the lithium metaborate fusion technique (Suhr and Ingamells, 1966; Ingamells, 1970).

For $^{40}Ar/^{39}Ar$ age-spectrum analysis (sites L3, L6, L9, and L21) 0.5-g samples were sealed in fused silica vials in air and irradiated in the U.S. Geological Survey TRIGA reactor for 40 MW hr where they received a total neutron dose of about 4×10^{18} nvt. Details of the irradiation procedure, the reactor flux characteristics, the flux monitor mineral, the corrections for interfering Ca-derived and K-derived argon isotopes, and the methods of calculation are described in Dalrymple and Lanphere (1971). $^{40}Ar/^{39}Ar$ age spectra were obtained by procedures described by Lanphere and Dalrymple (1971).

The oriented cores were cut into specimens 2.28 cm long, and the natural remanent magnetization (NRM) of one specimen from each core was measured with a spinner magnetometer (Doell and Cox, 1965). To remove secondary components of NRM, two or more pilot specimens from each site were progressively demagnetized in alternating fields up to 800 Oe using equipment described by Doell and Cox (1967b). For each site, the optimum demagnetizing field selected was that above which systematic changes in remanent magnetization (RM) direction ceased to occur. One specimen from each of the remaining cores from each site was then demagnetized at the optimum field for that site. Thermal stability of NRM was tested using additional specimens from each site by heating to progressively higher temperatures in air and cooling in a magnetic field of 100 γ or less. The RM was measured at room temperature after each heating stage. To aid in identifying the magnetic minerals, strong-field induced magnetization as a function of temperature was determined for specimens from all sites using an automatic-recording magnetic thermobalance (Doell and Cox, 1967c).

AGE OF THE DIKES

The results of conventional K-Ar age analyses on 25 samples from 22 sites are given in Table 1. The most striking feature of these results is the apparent lack of consistency. Calculated ages range from 173 m.y. to 1,213 m.y. (Fig. 2). Not only do the apparent ages of individual dikes differ, but ages are discordant to all levels of sampling. For example, the apparent ages of four separate samples from one dike at site L27, two from the fine-grained border zone (a and b) and two from the coarser interior (c and d), vary from 186 m.y. to 388 m.y. The apparent ages of different phases from the same hand sample also show large discordances, as for samples from sites L3 (whole rock, 853 m.y.; plagioclase, 671 m.y.; pyroxene, 493 m.y.) and L6 (whole rock, 1,121 m.y.; plagioclase, 429 m.y.; pyroxene, 595 m.y.).

The apparent ages of dikes and sills that intrude the Paynesville Sandstone are relatively concordant, ranging only from 173 m.y. to 192 m.y. In contrast, only two dikes that intrude Precambrian crystalline rocks have apparent ages that fall within or near this range, L27b (186 m.y.) and L29 (193 m.y.). The apparent ages of dikes that intrude rocks of the Liberian age province appear to be generally older than those of dikes that intrude crystalline rocks of the Pan-African age province (Fig. 2). Although the statistical samples

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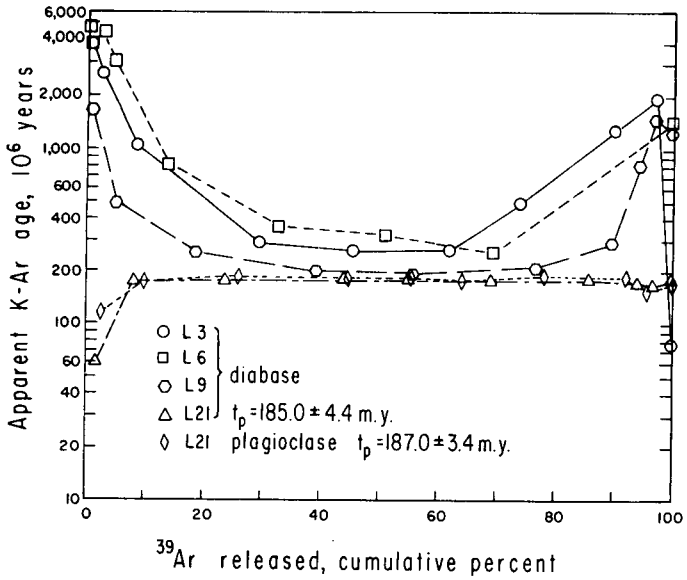


Figure 3. $^{40}Ar/^{39}Ar$ age spectra of whole-rock diabase and plagioclase from four Liberian sites. Samples L3, L6, and L9 have minima of 268 m.y., 265 m.y., and 203 m.y., respectively, and contain excess ^{40}Ar. Plateau ages (t_p) of L21 diabase and plagioclase probably represent emplacement age. Data for L21 plagioclase are from Dalrymple and Lanphere (1974); data for L3 and L6 are from Lanphere and Dalrymple (1971, samples 8L520 and 8L550).

are small, this difference is significant at the 88 percent level of confidence, indicating that the apparent age of individual diabase bodies may be partly a function of the age of the rocks into which they were emplaced.

On the basis of this apparent age pattern, it seems probable to us that the dikes and sills of western Liberia were formed at the same or nearly the same time but that those emplaced into Precambrian basement absorbed radiogenic ^{40}Ar from the surrounding crystalline rocks, whereas those emplaced into the Paynesville Sandstone did not. Accordingly, we interpret the ages from sites L19 through L25 as crystallization ages. Ages from all other sites are anomalous and must be considered maximum ages.

To further test the hypothesis that apparent ages significantly older than 190 m.y. are anomalous because of extraneous ^{40}Ar, $^{40}Ar/^{39}Ar$ age spectra were determined for four whole-rock diabase samples (sites L3, L6, L9, and L21) ranging in apparent age from 178 m.y. to 1,121 m.y. Plagioclase from L21 was also analyzed. The data for two of the whole-rock samples (L3 and L6) were reported by Lanphere and Dalrymple (1971) and for the plagioclase sample by Dalrymple and Lanphere (1974). The advantage of the $^{40}Ar/^{39}Ar$ age-spectrum technique (Merrihue and Turner, 1966; Turner, 1968, 1969, 1970; Fitch and others, 1969; Lanphere and Dalrymple, 1971; Brereton, 1972; Dalrymple and Lanphere, 1974) is that the ^{40}Ar that is not a product of the radioactive decay of ^{40}K within the rock would more likely inhabit different sites within the rock than would the ^{39}Ar, which is produced directly from ^{39}K by *n,p* reaction during irradiation in the reactor. This situation would result in a spectrum of different apparent ages from the different increments of gas released. From this age spectrum the true age, or some information about the true age, might be inferred. Conversely, for a rock that contains no extraneous ^{40}Ar and that has not been thermally disturbed, each increment of gas released as the sample is heated to progressively higher temperatures would be expected to have the same ratio of radiogenic ^{40}Ar to neutron-produced ^{39}Ar and, thus, the same apparent age. The ideal age spectrum for such a rock would be flat at a value corresponding to the crystallization age. Such age spectra are, in fact, characteristic of undisturbed terrestrial samples (Dalrymple and Lanphere, 1974).

TABLE 2. ANALYTICAL DATA FOR $^{40}Ar/^{39}Ar$ AGE SPECTRA OF WHOLE-ROCK LIBERIAN DIABASES L9 AND L21

Temp. (°C)	Measured ratios $^{40}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$*	$^{36}Ar/^{39}Ar$	^{39}Ar (% of total)	$^{40}Ar_{rad}$ (%)	Apparent age (10^6 years)
L9						
400	332.4	6.25	0.694	0.7	38.4	1709 ± 62
500	46.24	6.10	0.0682	4.5	57.4	505 ± 12
600	15.27	6.13	0.00882	13.6	86.1	267 ± 4
700	10.91	7.49	0.00460	20.6	93.0	209 ± 3
780	11.24	6.51	0.00641	16.5	87.7	203 ± 4
860	11.61	3.883	0.00410	21.0	92.2	220 ± 3
940	16.50	5.47	0.00612	11.9	91.7	304 ± 5
1020	52.7	19.22	0.01503	5.4	94.5	863 ± 10
1100	114.1	117.7	0.0698	3.0	90.2	1554 ± 17
Fusion	96.1	108.6	0.0836	2.9	83.4	1298 ± 18
L21						
400	27.47	1.721	0.0835	2.0	10.6	62 ± 20
500	10.88	1.571	0.00758	6.2	80.5	181 ± 4
600	9.47	3.158	0.00325	15.6	92.5	181 ± 2
700	9.73	3.511	0.00302	20.3	93.7	188 ± 2
780	9.74	1.793	0.00277	11.1	93.0	187 ± 3
860	9.52	1.123	0.00226	14.0	93.9	185 ± 2
940	9.68	1.298	0.00241	16.8	93.7	187 ± 2
1020	9.66	5.44	0.00490	8.3	89.5	179 ± 3
1100	11.36	53.7	0.0254	2.8	72.0	175 ± 11
Fusion	12.90	59.3	0.0302	3.2	67.6	187 ± 9

Note: The recalculated "total fusion" ages for these samples, found by mathematically recombining the data from the temperature steps, are L9, 394 ± 12 m.y.; L21, 182 ± 4 m.y.
* Corrected for ^{37}Ar decay (half-life = 35.1 days).
† The ± figures are estimates of the analytical precision at the 68 percent confidence level calculated using the method of Dalrymple and Lanphere (1971). $\lambda = 5.305 \times 10^{-1}\ yr^{-1}$, J = 0.01151.

The results (Fig. 3; Table 2) show that samples from sites L3, L6, and L9, all of which intrude Precambrian crystalline rocks of the Liberian age province, have large amounts of radiogenic ^{40}Ar that is uncorrelated with the neutron-produced ^{39}Ar. For some gas increments, this results in apparent ages exceeding 3,000 m.y. and, for one step, an apparent age older than the age of the Earth. All three age spectra have a distinctive shape. The apparent ages decrease to a broad depression at intermediate amounts of ^{39}Ar released, then increase until all ^{39}Ar is lost. These age spectra are unlike any yet obtained on lunar rocks (for example, Turner, 1970) or on terrestrial samples (Fitch and others, 1969; Lanphere and Dalrymple, 1971; Brereton, 1972; Dalrymple and Lanphere, 1974) and may be typical for rocks that contain excess ^{40}Ar.[1] The three age spectra appear to follow a pattern that is related to the total amount of excess ^{40}Ar in the sample (Fig. 3). As the apparent whole-rock age decreases, the depression becomes broader and lower. The age spectrum for L3, which gives an apparent age of 853 m.y., is generally below the age spectrum for L6, which gives an apparent whole-rock age of 1,121 m.y. These age spectra have similar minima of 265 m.y. (L6) and 268 m.y. (L3). The age spectrum for L9, which gives an apparent whole-rock age of 401 m.y., is below those for L3 and L6, has a broader depression, and reaches a minimum of 203 m.y.

The age spectra of the whole-rock diabase and the plagioclase from diabase L21,which is a sill that intrudes the Paynesville Sandstone, are essentially flat except for the first and, for the plagioclase, last few temperature increments. The remaining gas fractions form well-defined plateaus characteristic of essentially undisturbed terrestrial samples (Dalrymple and Lanphere, 1974). The means of plateau increments are 183.6 ± 4.5 m.y. and 186.5 ± 3.4 m.y. for the whole-rock diabase and plagioclase, respectively. If the apparent ages of the plateau increments are weighted according to the inverse of their estimated variance, the corresponding plateau values (t_p) are 185.0 ± 4.4 m.y. and 187.0 ± 3.4 m.y. The weighted means are preferred because the weighting takes into account the quality of each individual measurement.

The data for samples from sites L21 and L3 are plotted on $^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$ isochron diagrams in Figure 4. The ad-

[1] Similar age spectra have now been found for some ultramafic xenoliths that also contain excess ^{40}Ar (Kaneoka, 1974).

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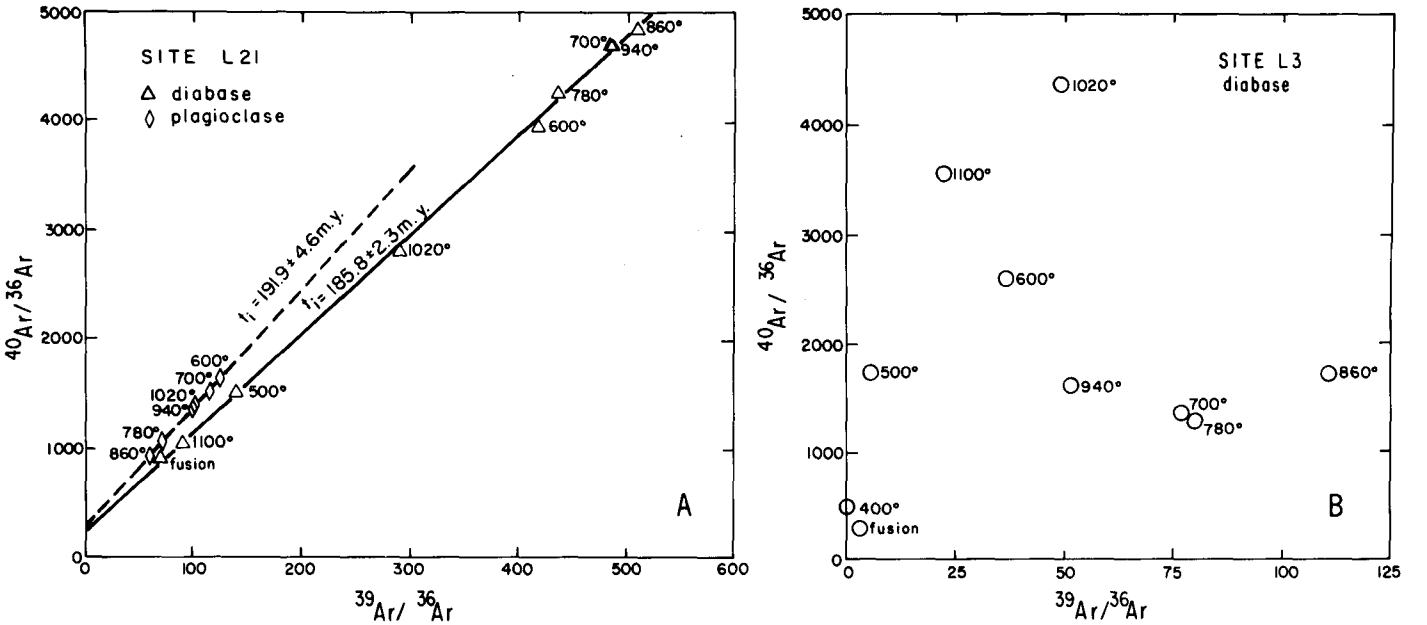


Figure 4. $^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$ isochron diagrams for whole-rock diabase and plagioclase from Liberian sites L21 (A) and L3 (B). Argon isotope ratios have been corrected for K-derived ^{40}Ar and Ca-derived ^{36}Ar and ^{39}Ar. Only plateau points are used for site L21 isochrons; indicated isochron ages (t_i) probably reflect emplacement age. Data for site L3 do not plot on an isochron because sample contains excess ^{40}Ar.

vantage of this treatment is that it is not necessary to assume that nonradiogenic argon has the isotopic composition of atmospheric argon. The age of the sample is given by the slope of the isochron calculated using a two-error, least-squares equation that allows for correlated errors (York, 1969). A discussion of the application of this data-reduction technique to $^{40}Ar/^{39}Ar$ incremental heating experiments is given by Dalrymple and Lanphere (1974). The data from both the plagioclase and the whole-rock diabase from site L21 fall on isochrons with indicated ages of 191.9 ± 4.6 m.y. and 185.8 ± 2.3 m.y., respectively (Fig. 4A). The $^{40}Ar/^{36}Ar$ intercepts of the isochrons are 264 ± 26 m.y. (plagioclase) and 276 ± 19 m.y. (whole rock), neither of which is significantly different from the atmospheric $^{40}Ar/^{36}Ar$ ratio of 296 at the 95 percent level of confidence. In contrast, the data from site L3 show no correlation and clearly do not fit an isochron (Fig. 4B). This is further evidence that a significant amount of the ^{40}Ar in this sample is not correlated with the ^{40}K but is excess ^{40}Ar. Although not shown, the data from sites L6 and L9 have similar scatter and do not fit isochrons.

Because the calculations of the isochron ages involve fewer assumptions about the composition of nonradiogenic argon, they are considered the best values for the age of site L21. The conventional K-Ar and $^{40}Ar/^{39}Ar$ incremental heating data demonstrate that sample L21 does not contain extraneous ^{40}Ar in detectable amounts and that its crystallization age is about 189 m.y.

The $^{40}Ar/^{39}Ar$ data on sites L3, L6, L9, and L21 substantiate the hypothesis that ages from dikes and sills that intrude the Paynesville Sandstone represent crystallization ages, whereas ages from dikes that intrude Precambrian crystalline rocks are only maximum ages because of extraneous ^{40}Ar. The data further indicate that the extraneous ^{40}Ar in the latter dikes is not confined to a single mineral or site but generally pervades the rock. In the two samples from which coexisting phases were analyzed (L3 and L6), there is a general decrease in apparent age with decreasing K_2O content. This suggests that the extraneous ^{40}Ar is somewhat more highly concentrated relative to ^{40}K in the potassium-rich phases. In basalt and, presumably, diabase, these phases are usually among the last to crystallize and often are interstitial potassium feldspar (Dalrymple and Lanphere, 1969; Mankinen and Dalrymple, 1972). Even the pyroxenes, which contain less than 0.1 percent K_2O, give ages more than twice the crystallization age. This general decrease in apparent age with decreasing potassium content (and hence order of crystallization), and the fact that dikes that intrude Paleozoic sedimentary rocks do not give anomalous ages, suggests that the extraneous ^{40}Ar was not contained in the diabase magma but was absorbed from the surrounding Precambrian rocks during crystallization. This hypothesis is further strengthened by the data from site L27. The two L27 samples from the more rapidly cooled border zone (a and b) have younger apparent ages than the two L27 samples from the coarser interior (c and d). This is just the opposite of submarine pillow basalts that contain excess ^{40}Ar dissolved in the magma when they erupt (Dalrymple and Moore, 1968; Dymond, 1970). Xenoliths are uncommon in the diabase, and there is no petrographic evidence that the extraneous ^{40}Ar is introduced by xenoliths or xenocrysts. The available evidence indicates that the extraneous ^{40}Ar was absorbed by the diabase from the Precambrian basement rocks and that the diabase continued to incorporate the argon until crystallization was complete.

Previous workers (McDougall, 1961, 1963; Miller and Mussett, 1963; McDougall and Ruegg, 1966) have not reported excess argon in tholeiitic diabase intrusive bodies of Mesozoic and Paleozoic age from Tasmania, Antarctica, South Africa, England, and South America, but these diabase bodies generally intrude Paleozoic or Mesozoic sedimentary rocks. K-Ar ages on northwest-trending diabase dikes and on the Freetown mafic igneous complex, both of which intrude Precambrian crystalline rocks in Sierra Leone, range from 172 m.y. to 1,875 m.y. (Briden and others, 1971; Allen and others, 1967). N. J. Snelling (quoted in Briden and others, 1971) suggested that the oldest dates are more realistic and that the younger dates are the result of argon loss. Briden and others (1971) and Andrews-Jones (quoted in Briden and others, 1971) present geologic, K-Ar, and paleomagnetic evidence that the Freetown igneous complex and the dikes are Mesozoic; they reason that the older ages are caused by excess ^{40}Ar from the basement rocks. The Sierra Leone dikes (and perhaps the Freetown igneous complex) may be genetically related to the dikes in Liberia, and the causes of the anomalous ages are probably similar.

In summary, we conclude that all of the diabase dikes and sills of the central and coastal zones are approximately contemporaneous.

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TABLE 3. SITE LOCATION AND MAGNETIC DATA FOR LIBERIAN DIABASE DIKES AND SILLS

Site	N. lat	W. long	N	H	D	I	α_{95}	R	k	k_o	$J_o \pm$ s.d., x 10^3	T_c
DIKES THAT INTRUDE PRECAMBRIAN ROCKS OF LIBERIAN AGE PROVINCE												
L1*	6.94	10.76	10	---	----	----	----	---	---	1.3	120. ± 95.	574°
L2	7.34	11.14	10	100	348.3	- 4.0	2.7	9.9717	318.	13.	2.0 ± 0.3	558°
L3	7.24	11.05	10	100	334.1	7.6	2.8	9.9707	307.	12.	3.4 ± 0.8	565°
L4	7.23	11.03	10	100	332.4	13.5	2.2	9.9807	466.	3.4	3.9 ± 0.9	563°
L5†	7.07	10.88	10	200	(261.2)	(3.9)	26.0	7.9611	4.4	3.0	200. ±175.	576°
L6*	7.01	10.78	10	---	----	----	----	---	---	1.7	60. ± 9.	572°
L7	6.69	10.23	10	100	328.7	-24.4	3.3	9.9592	221.	100.	2.7 ± 0.6	576°
L8	6.59	10.08	(Unoriented sample collected for dating only)									
L9	6.69	10.22	9	50	336.9	5.9	2.6	8.9806	411.	25.	2.5 ± 0.7	576°
L10†	6.76	10.09	10	100	(328.1)	(4.9)	26.0	7.9659	4.4	1.5	8.5 ± 4.2	575°
L11	6.83	9.87	10	200	330.0	15.5	1.8	9.9882	764.	3.1	5.0 ± 2.2	572°
L12	6.84	9.85	8	200	339.9	15.6	4.5	7.9535	151.	5.6	6.7 ± 3.9	578°
L13	6.85	9.84	10	200	339.6	27.3	2.9	9.9688	289.	18.	6.1 ± 1.1	576°
L14	6.88	9.73	9	200	356.5	4.9	4.6	8.9377	128.	2.5	42. ± 19.	562°
L15	6.79	9.95	9	100	321.4	-17.3	4.8	8.9306	115.	114.	1.5 ± 0.2	576°
L16	6.78	9.97	10	50	347.9	-21.5	3.4	9.9562	206.	247.	1.8 ± 0.5	567°
L17	6.78	9.99	9	100	340.1	5.6	4.4	8.9422	138.	38.	5.8 ± 0.8	576°
DIKES AND SILLS THAT INTRUDE PAYNESVILLE SANDSTONE AND PRECAMBRIAN ROCKS OF PAN-AFRICAN AGE PROVINCE												
L18	6.34	10.73	10	25	327.1	27.6	8.6	9.7236	32.	27.	1.5 ± 0.5	479°
L19*	6.31	10.69	10	---	---	---	---	---	---	1.5	53. ± 55.	549°
L20	6.27	10.72	8	100	348.0	- 6.7	5.9	7.9223	90.	4.3	0.69± 0.41	498°
L21	6.23	10.62	10	100	0.9	-14.8	2.3	9.9798	445.	2.6	1.4 ± 0.6	540°
L22	6.20	10.55	10	100	352.5	-12.3	4.7	9.9166	108.	2.1	1.1 ± 0.7	560°
L23	6.23	10.50	10	100	353.1	- 6.9	3.0	9.9648	255.	5.3	1.5 ± 0.9	553°
L24*	6.28	10.77	10	---	---	---	---	---	---	1.4	36. ± 46.	539°
L25	6.32	10.82	10	50	349.4	-24.7	1.6	9.9908	978.	334.	0.68± 0.18	546°
L26	6.25	10.31	10	100	141.9	-21.6	2.1	9.9835	547.	9.4	3.2 ± 0.7	576°
L27	6.12	10.18	16	100	159.4	36.5	3.1	15.8941	142.	25.	1.1 ± 0.3	576°
L28	6.12	10.13	10	50	155.2	- 3.2	2.7	9.9725	327.	33.	2.5 ± 0.5	570°
L29	5.86	10.05	9	100	341.1	- 8.8	4.3	8.9454	147.	1.5	1.8 ± 1.2	558°
L30	5.99	10.02	10	200	158.1	- 5.1	2.3	9.9800	450.	12.	6.0 ± 3.2	570°
L31	6.28	10.33	10	50	135.7	- 5.7	3.1	9.9631	244.	138.	1.3 ± 0.2	575°
L32	6.30	10.32	10	200	164.9	-10.8	3.0	9.9656	262.	12.	1.3 ± 1.1	573°

Explanation: Locations; N lat and W long, degrees. N = number of specimens. H = peak alternating field used for demagnetization. D = mean RM declination, degrees E of N. I = mean RM inclination, degrees positive downward. α_{95} = semiangle of 95 percent confidence cone. R = vector resultant of N unit directions. k = precision parameter (Fisher, 1953). (D, I, α_{95}, R, and k are for RM directions after a.f. demagnetization.) k_o = precision parameter for mean NRM directions (before demagnetization). J_o = mean NRM intensity in e.m.u./cm^3. T_c = Curie temperature in degrees C.

*Trial demagnetizations showed no convergences of specimen RM directions.

†Precision of mean too low for use in average paleomagnetic pole computations.

Reliable K-Ar ages fall within the range of 173 m.y. to 192 m.y., making the intrusions earliest Jurassic in age (Harland and others, 1964). This range in calculated ages is greater by about a factor of two than that expected from analytical uncertainty and may represent small differences in time of emplacement.

PALEOMAGNETISM OF THE DIKES

The paleomagnetic results for sites from which oriented cores were collected are summarized in Table 3. Most of the sites have been thoroughly overprinted with strong isothermal remanent magnetizations (IRM) resulting from lightning. The success of alternating field (a.f.) demagnetization in preferentially removing the IRM and recovering the original homogeneous NRM directions in the diabases may be seen in Table 3 by comparing values of k_o, the precision parameter of directions before demagnetization, with corresponding values of k, the precision parameter after demagnetization. In all but seven instances, the dispersion of directions decreased significantly, usually markedly, after demagnetization. In four of the seven (sites L1, L6, L19, and L24), progressive a.f. de-

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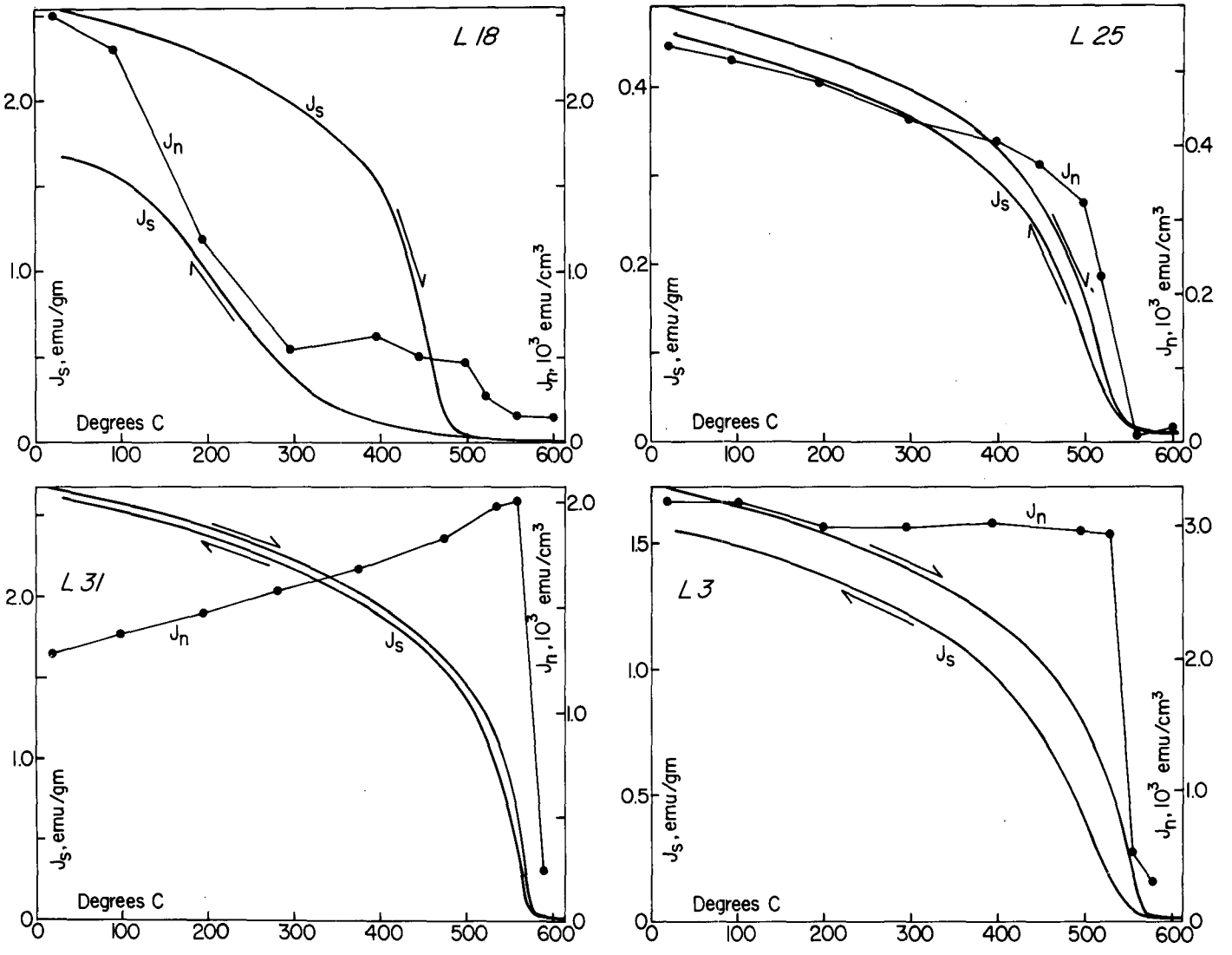


Figure 5. Thermal behavior of natural remanent magnetization (J_n) and strong-field induced magnetization (J_s) in specimens from four sites in Liberian diabase. Measurements of J_n were made at room temperature. J_s was measured on an automatic recording thermobalance; arrows indicate heating and cooling curves. All samples were heated in air. Site numbers are indicated on diagrams.

magnetization of several pilot specimens failed to produce convergence of the initially scattered NRM directions, and no further demagnetization was done. In two more of the seven (sites L5 and L10), progressive demagnetization produced some convergence of RM directions, but the dispersion of directions after demagnetization of all specimens was still too great to provide a paleomagnetically useful result. For site L18, the initially moderate dispersion decreased only slightly after demagnetization. In summary, for 25 out of 31 sites (including site L18), the semiangle of the cone of 95 percent confidence centered on the mean RM direction (α_{95}) is less than 9° after a.f. demagnetization (Table 3), and these data are paleomagnetically useful.

For most sites, the primary NRM (that is, homogeneous in direction) of these diabases was found to be highly stable against a.f. demagnetization. The mean destructive field (m.d.f.) has been defined as the peak alternating field required to decrease the intensity of remanent magnetization by one-half (Park and Irving, 1970). For specimens carrying little or no secondary NRM, m.d.f. values range from 100 Oe to 400 Oe, and average roughly 200 Oe.

The thermal stability of NRM in representative specimens from all sites was investigated by progressive heating as described above. Typical results for four sites are shown in Figure 5, in which thermal destruction of NRM (J_n) is compared with variation with temperature of strong-field induced magnetization (J_s). Diabase from site L18 showed unique thermomagnetic properties in that the magnetic mineral is highly altered by heating, as indicated by the J_s curves, and the NRM is unstable after heating to 300°C. Of all the sites considered to carry paleomagnetically useful NRM directions, this one showed the poorest response to a.f. demagnetization (Table 3). For the other three sites illustrated in Figure 5, the curves of thermal destruction of NRM show two distinct segments. The low temperature segments may have either positive or negative slopes and are associated with removal of secondary components of NRM consisting of viscous and lightning-produced RMs. The high-temperature segments represent removal of primary NRM and are usually restricted to within 30° to 50°C below the Curie temperatures indicated by the J_s curves. With the exception of site L18, the Curie temperatures range from 503° to 578°C, and in no case did any part of the NRM survive heating above the bulk Curie temperatures indicated by the J_s curves. Hence it appears that the primary NRM is carried by the mineral represented by the J_s curves and that this is a titanium-poor titanomagnetite containing 0 to 12 mole percent of the ulvöspinel component.

The high-temperature segments of the NRM decay curves are relatively much smaller or are entirely missing for the sites that did not yield paleomagnetically useful data, probably indicating that most or all of the primary NRM was destroyed by the very large fields associated with lightning currents. Nevertheless, an attempt

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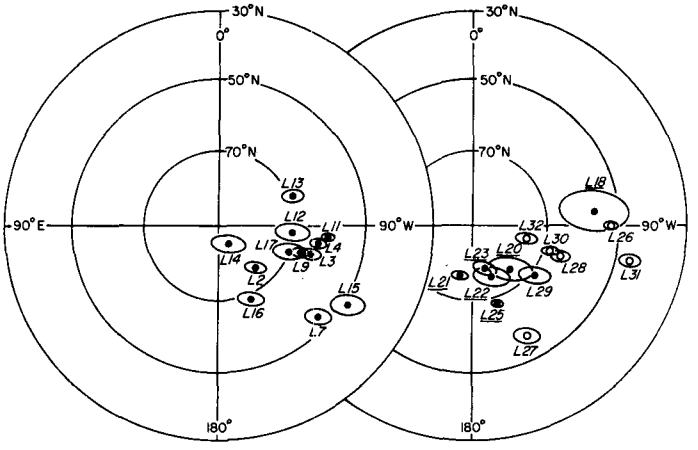


Figure 6. Virtual geomagnetic poles corresponding to mean NRM directions after a.f. demagnetization at 25 sites in Liberian diabase. *Left*, dikes that intrude rocks of the Liberian age province. *Right*, dikes and sills that intrude Paynesville Sandstone and rocks of Pan-African age province. Closed circles = normal polarity; open circles = reversed polarity. Underlined site numbers indicate VGPs obtained from diabase that intrudes Paynesville Sandstone. Ovals around poles are transformed 95 percent confidence circles (Table 4). Equal-area projection on northern 60° of northern hemisphere.

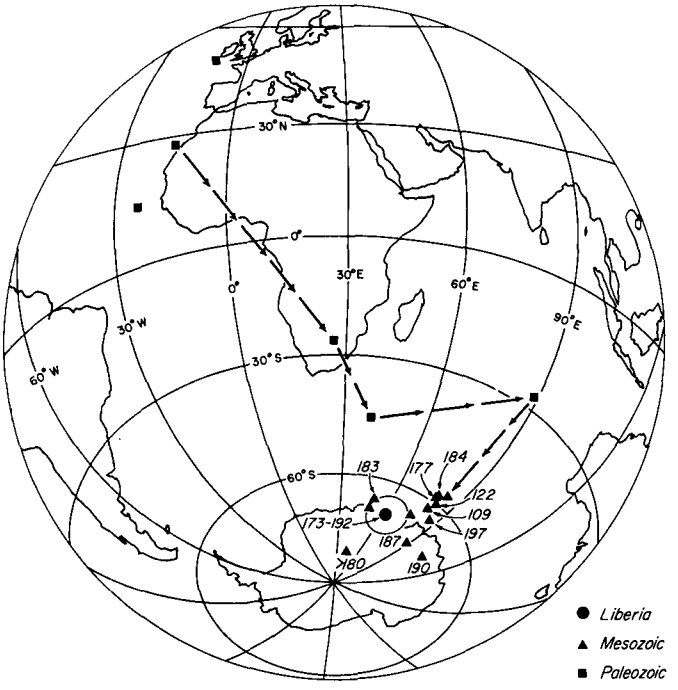


Figure 7. Polar wandering path for Africa during Paleozoic and Mesozoic time. Paleomagnetic pole from Liberian diabase is shown with 95 percent confidence circle. Radiometric ages for paleomagnetic poles are given in millions of years where available. Most data are from review of McElhinny and others (1968) with additional Mesozoic data from Briden and others (1971), Hailwood and Mitchell (1971), and Fitch and Miller (1971).

was made to use partial thermal demagnetization to recover the primary NRM directions for sites L10 and L19. A complete set of specimens from each site was twice heated, first to 500°C and then to 550°C, and cooled each time in null magnetic field to room temperature, but in no case was the precision parameter k greater than 4 after heating; that is, the decrease in dispersion of directions was negligible.

For each site that yielded paleomagnetically reliable NRM directions, a virtual geomagnetic pole (VGP) (Cox and Doell, 1960) with its associated 95 percent oval of confidence has been calculated from the data in Table 3 and is illustrated in Figure 6. Of the instances mentioned above in which more than one sampling site may be located on the same dike, only the pair L3 and L4 are represented in Figure 6. The VGPs for these two sites are significantly different, and we conclude that they are not in the same dike because exposures are not continuous. Each VGP in Figure 6, therefore, probably represents a separate point in time. These VGPs form a homogeneous group; that is, there are no sites that have anomalous NRM directions. Dikes that intrude rocks of the Liberian age province do not show systematically different VGP positions from dikes and sills that intrude the Paynesville Sandstone and rocks of the Pan-African age province. The only reversely magnetized dikes occur in a somewhat restricted area within the Pan-African age province east of the area in which sedimentary rocks crop out (Figs. 1 and 6). Although these reversed VGPs appear to form a group slightly different from the group of VGPs of rocks intruding the Paynesville Sandstone, the difference is not significant at the 95 percent confidence level. Furthermore, the group of reversed VGPs is not at all different from the group of all normal VGPs (Fig. 6), which demonstrates that no systematic secondary component of NRM remained in the diabase specimens after a.f. demagnetization and that the primary NRM in the diabases was almost certainly acquired during their initial cooling after intrusion. The group of VGPs obtained from diabase intruding the Paynesville Sandstone, with the exception of site L18, forms a tighter cluster than the other VGPs. This suggests the possibility that the total time of diabase intrusion may have been somewhat greater than the time indicated by the dating of diabase in the Paynesville Sandstone.

Mean paleomagnetic poles have been calculated from the data in Table 3 and are given in Table 4. The angular dispersions of the various groups of VGPs shown in Table 4 are all similar, and the overall dispersion is 14.6° with upper and lower 95 percent confidence limits of 18.1° and 12.2° (Cox, 1969). McElhinny and others (1968) have cited a range of 5.5° to 12.6° for the angular dispersions of paleomagnetic field directions in six other studies of African Mesozoic rocks. Assuming azimuthal symmetry of virtual poles about their means and using paleolatitudes calculated from the mean inclinations cited by McElhinny and others (1968), these values are equivalent to 6.4° to 11.7° for the range of angular dispersions of virtual poles (Cox, 1970). The dispersion of the Liberian VGPs is not significantly greater than the maximum of this range at 95 percent confidence. (The angular dispersion of the 44 VGPs used to compute a mean latest Triassic to earliest Jurassic paleomagnetic pole for northwest Africa, described below, is 13.4°; this includes the Liberian data.) A summary of secular variation for

TABLE 4. MEAN PALEOMAGNETIC POLES FOR LIBERIAN DIABASE

	N	R	k	δ	α_{95}	N. lat	E. long
Dikes that intrude rocks of Liberian age province:	12	11.7149	39.	13.1	7.1	67.5	246.7
Dikes and sills that intrude Paynesville Sandstone and Precambrian rocks of Pan-African age province:							
Normal polarity	7	6.8274	35.	13.8	10.4	74.4	221.1
Reversed polarity	6	5.8180	27.	15.5	13.0	62.0	249.4
Normal and reversed	13	12.5240	25.	16.2	8.4	69.3	238.1
All sites	25	24.2267	31.	14.6	5.3	68.5	242.4

Explanation: δ = r.m.s. angular dispersion of virtual poles about mean pole, N = number of sites; other symbols as in Table 4.

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TABLE 5. PALEOMAGNETIC POLES USED IN CONTINENTAL RECONSTRUCTION

Locality and rock type	Number of samples/sites/ and rock units	N. lat	E. long	α_{95}	References	K-Ar ages, m.y.	References
Eastern North America							
Nova Scotia							
North Mountain basalt, upper lavas	21/16/?	71°	149°	4.7°	(Carmichael and Palmer, 1968)	217	(same)
North Mountain basalt, lower lavas	17/11/?	67	67	5.5		178, 196, 204	
Shelburne dike	11/2/1	69	98	4.5	(Larochelle and Wanless, 1966)	197	(same)
Connecticut							
Talcott Basalt	51/10/2	51	78	4.0	(de Boer, 1968)		
Holyoke Basalt	81/10/2	60	83	3.0		193, 161, 176, 197, 201	(de Boer, 1968) (Armstrong and Besancon, 1970)
Hampden Basalt	76/14/9	73	92	4.0		195	(Armstrong and Besancon, 1970)
West Rock dike	18/3/1	63	89	4.0		190, 198, 201	(Armstrong and Besancon, 1970)
Mount Carmel sill	33/5/1	61	87	4.0			
Cheshire dikes	41/?/2	76	103	4.0			
Massachusetts							
Holyoke Basalt and Granby Tuff of Newark Group	16/5/5?	55.5	91.1	12.0	*(Irving and Banks, 1961)		
New Jersey							
Watchung Basalt of Newark Group	13/5/3	61.4	83.5	8.3	*(Opdyke, 1961)		
Palisades sill and other intrusive rocks	17/6/5	64.2	112.9	8.6		190, 193, 202, 202 (Palisades) 192	(Erickson and Kulp, 1961), (Armstrong and Besancon, 1970)
Pennsylvania							
Gettysburg sill	57/13/1	65	103	8.3	(Beck, 1972)		
Yorkhaven sill	202/37/1	61	105.5	1.5			
Birdsboro sill	168/26/1	62	105	3.2			
Quakertown sill	50/9/1	66	98.5	4.3			
Mean pole for Eastern North America	N = 16	65.0	94.7	4.4			
Western Africa							
Morocco							
Draa Valley sills: MC 1	6/1/1	68.0°	215.0°	3.0°	Hailwood and Mitchell, 1971)		(same)
MC 2	6/1/1	64.5	215.0	6.0			
MC 3	6/1/1	64.5	225.5	3.0			
MC 4	6/1/1	65.5	225.0	10.0			
MC 5	6/1/1	71.5	203.5	8.0			
MC 6	6/1/1	64.5	200.0	12.0			
MC 7	6/1/1	57.5	228.0	3.0			
MC 8	6/1/1	59.0	235.5	2.0			
MC 9	6/1/1	58.5	241.5	7.0			
MC10	6/1/1	66.0	234.5	3.0			
MC11	6/1/1	64.0	239.5	5.0			
MC12	6/1/1	70.5	260.5	6.0			
MC13	6/1/1	63.0	241.0	4.0			
MC14	6/1/1	71.5	222.5	2.0		180, 183	
MC15	6/1/1	61.5	243.0	7.0		183, 186	
MC16	6/1/1	63.5	248.0	4.0		181, 183	
Foum-Zguid dike	27/5/1	58.0	259.0	4.0		182, 182, 186, 187	
Sierra Leone							
Freetown complex, normal polarity	7/7/1	78.2	121.3	5.5	*(Briden and others, 1971)	165, 172, 194	(same)
Freetown complex, reversed polarity	3/3/1	87.0	154.5	8.4			
Liberia							
(Virtual poles for 25 dikes and sills calculated from data in Table 4 of this paper.)							
Mean pole for Western Africa	N = 44	68.3	237.2	3.6			

Paleomagnetic poles recalculated from data in original reference.

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Pliocene and Pleistocene time represented by angular dispersions of VGPs from many series of lava flows has recently been illustrated by Doell and Dalrymple (1973). The paleomagnetic latitude of the Liberian diabases is 0°, and hence the comparable dispersion data cited by Doell and Dalrymple are from the Galapagos Islands: 13.1° for 147 lavas of Brunhes age and 18.5° for 36 lavas of Matuyama age. The dispersion found for the Liberian diabases falls between these values and is not significantly different from either, suggesting that the full range of secular variation is represented in the Liberian data.

The mean paleomagnetic pole position for all Liberian diabases is illustrated in Figure 7 for comparison with other Paleozoic and Mesozoic paleomagnetic poles from the African continent. It is apparent from this figure that during the interval from 197 to 109 m.y. no discernible polar wandering occurred relative to Africa, a fact already pointed out (see, for example, Gough and others, 1964; Grommé and others, 1967; Briden, 1967; McElhinny and others, 1968). The youngest Paleozoic paleomagnetic pole from Africa (pole B6 of McElhinny and others, 1968) is Early Permian and is 30° removed from the group of Mesozoic poles (Fig. 7). Thus, for example, if an appreciable number of the Liberian diabase dikes were as old as 250 m.y., this would be indicated by much greater dispersion, and probably a more nearly linear or bimodal distribution, of the VGPs than is observed.

INTERCONTINENTAL COMPARISON OF PALEOMAGNETIC POLES

Some workers have examined the behavior of paleomagnetic poles for circum-Atlantic continents when these continental plates are restored to some predrift configuration (see, for example, Wells and Verhoogen, 1967; Creer and others, 1969; Veldkamp and others, 1971; Phillips and Forsyth, 1972; Roy, 1972). In most cases, Paleozoic and early Mesozoic paleomagnetic poles, which are widely separated with the continents in their present positions, converge when the continental plates are joined to form a single plate. We have performed a similar test, comparing only paleomagnetic data of a restricted age range (approximately 170 to 200 m.y.) from eastern North America and northwestern Africa, including the Liberia results in this study. The North American paleomagnetic localities are in basaltic igneous rocks distributed from Nova Scotia to Pennsylvania; the African localities are in igneous rocks in Morocco, Sierra Leone, and Liberia. An important element of this test is the likelihood of increased precision due to the fact that, after Africa is joined to North America, the various sampling localities are separated by distances that are small in comparison to the sizes of the continental plates.

The data we used and the paleomagnetic poles derived from them are summarized in Table 5. In calculating a composite paleomagnetic pole by averaging a number of poles, it is important that all poles (or virtual poles) of the group represent similar spans of time, that is, that each integrates the same amplitude of geomagnetic secular variation. Because the data in the literature are not uniform in this respect, compromises were made. For the mean pole for northwest Africa, we included the 25 virtual poles from the Liberian dikes, together with 16 virtual poles from single sills and one mean pole from a single large dike (the Foum-Zguid dike), all in Morocco (Hailwood and Mitchell, 1971). In order to include the data from the Freetown igneous complex in Sierra Leone (Briden and others, 1971), we computed two mean poles, one each from the normally and reversely magnetized sites. Clearly, the three poles from the large dike and the Freetown complex represent more time and thus averaged more secular variation than the others and hence may have been given somewhat too little weight in the mean pole for Africa. The same problem exists for some of the North American data in that the sites in the North Mountain basalt of Nova Scotia and all of the Massachusetts and New Jersey sites may also have been given too little weight in the mean pole. A similar Late Triassic mean pole for North America has been calculated by Beck (1972), weighting the data somewhat differently; his result (lat 66° N., long 95.5° E., α_{95} = 3.8°) differs by less than 2° of arc from that given in Table 5.

Three different continental reconstructions have been tested and are illustrated in Figure 8. In every case the convergence of the North American and African poles is striking. The well-known predrift reconstruction of Bullard and others (1965), obtained by minimizing the misfit of the 500-fathom isobaths on the opposite margins of the Atlantic, is shown in Figure 8B. With the continental plates in their present positions, the great-circle distance between the two latest Triassic and earliest Jurassic poles is 44°; after rotation, this distance is reduced to only 3°. Dietz and others (1970) pointed out a difficulty inherent in the reconstruction by Bullard and others (1965), namely that part of the African craton overlaps the Bahamas platform and part of Florida. An alternate juxtaposition of Africa and North America is illustrated by Dietz and his colleagues (1970) but the rotation parameters were not given. A reconstruction that partially relieves this misfit is that of LePichon and Fox (1971), who obtained it by minimizing the mismatch between features presumed to represent fracture zones created during the early separation of Africa and North America. These features are the Guinea fracture zone and the Bahama escarpment, and the Cape Verde archipelago and the Cape Fear arch. In the reconstruction by LePichon and Fox (shown in Fig. 8C), the angular difference between the two poles is further reduced to 1.8°, an improvement that is not, however, statistically significant.

Another reconstruction of the circum-Atlantic plates has been suggested by Phillips and Forsyth (1972), who sought to minimize the dispersion of paleomagnetic poles from all the plates for Carboniferous through Tertiary time. Their reconstruction produced a slight reduction in dispersion of the Triassic poles used by them; the precision parameter k of the poles was 52.9 as compared with a k of 41.0 for the reconstruction by Bullard and others (1965). In the Phillips and Forsyth reconstruction (Fig. 8D), the coincidence of the African and North American poles for latest Triassic to earliest Jurassic time is clearly not as close as in the other reconstructions. The angular separation of the poles is 7°, although this difference is not significant at the 95 percent confidence level. We conclude that the conjunction of Africa and North America proposed by LePichon and Fox (1971) best satisfies both the geologic and the paleomagnetic evidence.

INITIATION OF SEPARATION OF AFRICA AND NORTH AMERICA

The initial event in the breaking apart of ancient Pangaea and the formation of the present Atlantic Ocean is considered by many authors to be the separation of the African and North American plates (Dietz and Holden, 1970; Pitman and Talwani, 1972; Phillips and Forsyth, 1972; Larson and Pitman, 1972; Smith and others, 1973). The recent consensus appears to be that this separation began between 180 and 200 m.y. ago, based on the following three lines of evidence:

1. Basaltic igneous activity represented by lava and intrusive rocks in the eastern Appalachians (the Late Triassic Newark Group) may either mark the initiation of rifting or predate it by as much as 20 m.y. (Vine and Hess, 1970). With exceptions discussed

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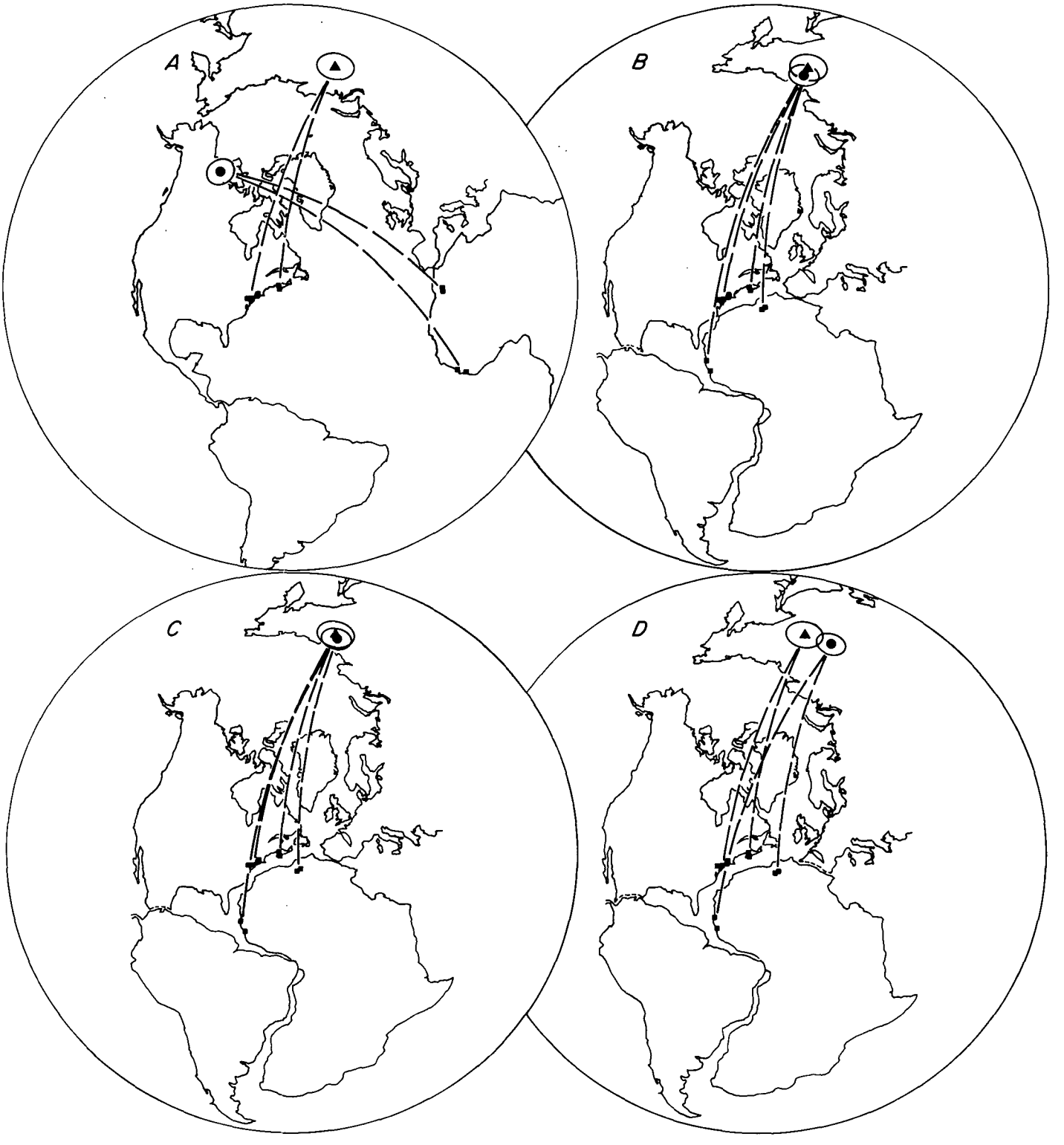


Figure 8. Comparisons of latest Triassic to earliest Jurassic paleomagnetic poles for Africa and North America with various reconstructions of prerift position of continents. Squares are sampling localities, closed circle is African pole, triangle is North American pole. Ninety-five-percent confidence circles are shown around both poles. For pole coordinates and data from which they were obtained, see Table 6. Dashed lines are paleomeridians connecting poles with sampling localities. Map is equal-area projection centered on pole at 45° N., 300° E., with North America left fixed. A, Continents in their present positions. B, Reconstruction of Bullard and others (1965). C, Reconstruction of LePichon and Fox (1971). D, Alternate reconstruction of Phillips and Forsyth (1972).

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below, K-Ar ages of these rocks range from 161 to 217 m.y., with most ages occurring within the interval from 184 to 202 m.y. (Reesman and others, 1973; and data cited in Table 6).

2. The zone of the ocean floor bordering the continental shelf of North America within which linear magnetic anomalies are subdued or absent (the "quiet zone") has a counterpart off the west coast of North Africa (Heirtzler and Hayes, 1967). Pitman and Talwani (1972) summarized the various hypotheses for the origin of these quiet zones and showed that their seaward margins can be rotated into congruence and hence are probably isochrons. If this is so, the quiet zones are probably due to a period of sea-floor spreading during which the geomagnetic field had a nearly constant polarity. McElhinny and Burek (1971) adopted this explanation and correlated the quiet zones with a Late Triassic to Late Jurassic (approximately 200 to 150 m.y. ago) period of dominantly normal polarity, which they established using paleomagnetic data from continental rocks. McElhinny and Burek (1971) thus inferred an age of about 200 m.y. for the edge of the continental shelf, that is, for the inception of active rifting.

3. Paleontologic ages of sediments overlying basaltic basement rocks in holes 9, 100, and 105 of the Deep Sea Drilling Project have allowed an approximate estimate to be made of the sea-floor spreading rate in the northwestern Atlantic Ocean during the interval from 155 to 120 m.y. If these ages and this rate are extrapolated westward to the continental rise, 175 to 180 m.y. is obtained for the time of the beginning of opening of the Atlantic (Scientific Staff, 1970).

The coincidence of latest Triassic to earliest Jurassic paleomagnetic poles from eastern North America and northwestern Africa after the two continental plates are joined (Fig. 8) shows that these two continents could not have separated very far prior to that time (see also Larson and LaFountain, 1970). The earliest limit for the beginning of separation can be estimated from the existing K-Ar ages and is approximately 180 m.y. (see Table 6). Hence if the magnetic quiet zones adjacent to the continental margins are due to generation of oceanic crust during a long period of constant geomagnetic polarity, this period cannot be the Kiaman interval (Permian) of reversed polarity, as originally postulated (Heirtzler and Hayes, 1967; Emery and others, 1970), but is almost certainly the Graham interval (Triassic and Jurassic) of normal polarity (McElhinny and Burek, 1971; Pitman and Talwani, 1972).

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